Statistics 280 — Practice Midterm 1A — Solutions

Point values per part match the exam. Normal-table values follow the table at the back of the exam; answers within rounding of these are fine.

Ex. 1 (30 points)

a (8 points)

Families choose whether to apply, so applicants differ from non-applicants: more motivated students, more involved parents, often higher socioeconomic status. These factors are associated with the treatment (attending the charter school) and affect the outcome (test scores), so they are confounders. The table shows it directly: the lottery *losers*, who never attended, score 0.20 − (−0.10) = 0.30 standard units above non-applicants. Most of the 0.45 gap is due to who applies, not to the school. The bias is **in favor of** the charter school.

b (8 points)

Compare winners to losers: 0.35 − 0.20 = **0.15 standard units**. Both groups applied, and a random lottery decided who got a seat, so the groups should be similar in motivation, family background, etc. It is a **natural experiment**: the researchers did not assign treatment, and the subjects did not choose it either; an external random mechanism (the lottery) did. (Saying it is "essentially a randomized experiment among applicants" earns credit with that explanation.)

c (7 points)

The 150 decliners chose not to attend; they may differ from those who accepted (less motivation, transportation problems, or better alternatives). Dropping them leaves a self-selected treatment group, so the comparison is observational, even inside a randomized study. This is the issue from the **Clofibrate trial** (adherers vs. non-adherers) and from the HIP screening trial (examined vs. refused). The fair comparison is all 600 winners vs. all 1,400 losers.

d (7 points)

The estimate applies to applicants. Non-applicant families may benefit less (or more), so the effect need not generalize. Also, expanding the school to 10,000 students changes the school itself (class sizes, teachers, peer groups), so the treatment would not be the same.

Ex. 2 (30 points)

a (6 points)

Hospital A: good 196/200 = 98%, poor 640/800 = 80%, overall 836/1000 = **83.6%**.
Hospital B: good 776/800 = 97%, poor 150/200 = 75%, overall 926/1000 = **92.6%**.

b (7 points)

Choose **Hospital A**: it has the higher survival rate for both kinds of patient. The overall rate is a weighted average of the condition-specific rates, with weights given by each hospital's patient mix.

A treats mostly poor-condition patients (80%), B mostly good-condition patients (80%). Patient condition is a confounder: associated with the hospital and with survival. This is Simpson's paradox.
Standardized: A: $0.5(98) + 0.5(80) = 89\%$; B: $0.5(97) + 0.5(75) = 86\%$. On the same patient mix, A is better.

c (10 points)
Let $x_1 \le \cdots \le x_8$. Each number is 12.5% of the list. The 25th percentile is the smallest number with at least 2 numbers at or below it, so $x_2 = 5$. Similarly the median is $x_4 = 7$ (need 4) and the 75th percentile is $x_6 = 11$ (need 6). Also $x_1 = 2$, $x_8 = 15$. Unknown: $5 \le x_3 \le 7$, $7 \le x_5 \le 11$, $11 \le x_7 \le 15$.
Mean 8.5 means the sum is 68, so $x_3 + x_5 + x_7 = 68 - (2 + 5 + 7 + 11 + 15) = 28$. Since 11 is the unique mode, it appears at least twice, so $x_5 = 11$ or $x_7 = 11$.

- $x_5 = x_7 = 11$: $x_3 = 6$. List 2, 5, 6, 7, 11, 11, 11, 15. 11 appears three times, everything else once. Valid.
- $x_5 = 11,\ x_7 \ge 12$: $x_3 = 17 - x_7 \le 5$, so $x_3 = 5, x_7 = 12$: 2, 5, 5, 7, 11, 11, 12, 15. Now 5 and 11 both appear twice. Not a unique mode.
- $x_7 = 11,\ x_5 \le 10$: $x_3 = 17 - x_5 \ge 7$, so $x_3 = 7, x_5 = 10$: 2, 5, 7, 7, 10, 11, 11, 15. 7 and 11 both twice. Not valid.

The only list is **2, 5, 6, 7, 11, 11, 11, 15**.

> Trap: the class definition of percentile, not "average the two middle numbers." The whole-number condition matters too: without it, 2, 5, 5.5, 7, 11, 11, 11.5, 15 would also work.

d (7 points)
The maximum SD is **5**, attained by any list that is half 0s and half 10s (e.g. 0, 10): the mean is 5 and every deviation is ± 5.
Proof: for $0 \le x \le 10$, $x(10 - x) \ge 0$, so $x^2 \le 10x$. Averaging over the list, $\overline{x^2} \le 10\bar{x}$. By the shortcut formula

$$SD^2 = \overline{x^2} - \bar{x}^2 \le 10\bar{x} - \bar{x}^2 = 25 - (\bar{x} - 5)^2 \le 25.$$

So $SD \le 5$. (W25 showed SD ≤ 10; that bound is true but not tight.)

Ex. 3 (35 points)

Block percentages (height × width): [0,10) 10%, [10,20) 20%, [20,25) 15%, [25,40) 30%, [40,60) 20%, [60,100) 0.125 × 40 = 5%. They sum to 100%.

a (5 points)
[15,20): $2 \times 5 = 10\%$; [20,25): 15%; [25,30): $2 \times 5 = 10\%$. Total $\mathbf{35\%}$.

b (8 points)
Cumulative: 10% at 10, 30% at 20, 45% at 25, 75% at 40. The median is in [25,40); we need 5% of that block's 30%: median $= 25 + 15 \cdot \frac{5}{30} =$ **27.5**.
Mean, with each block's values centered at its midpoint: $5(0.10) + 15(0.20) + 22.5(0.15) + 32.5(0.30) + 50(0.20) + 80(0.05) =$ **30.63**.
Mean > median because of the long right tail (the 60–100 block).

c (7 points)
25th percentile: 10% below 10, need 15% of the 20% block [10,20): $10 + 10 \cdot \frac{15}{20} = 17.5$. 75th percentile: exactly 75% is below 40, so it is 40. IQR $= 40 - 17.5 =$ **22.5**.
90th percentile: need 15% of the 20% block [40,60): $40 + 20 \cdot \frac{15}{20} =$ **55**.

d (7 points)
(i) $\sum |x_i - 2c|$ is minimized when $2c$ is the median: $2c = 27.5$, $c = 13.75$.
(ii) $\sum (x_i - (c + 10))^2$ is minimized when $c + 10$ is the mean: $c = 30.625 - 10 =$ **20.63**.

e (8 points)
$y = 100 - 2x$ is decreasing, so it reverses the order but keeps the middle in the middle. Median $= 100 - 2(27.5) =$ **45**. Mean $= 100 - 2(30.625) =$ **38.75**. The 25th percentile of y comes from the 75th of x: $100 - 80 = 20$; the 75th of y from the 25th of x: $100 - 35 = 65$. IQR $= 45 = 2 \times 22.5$.
SD(y) $= |-2| \cdot$ SD(x) $= 2$ SD(x). Adding 100 changes nothing; multiplying by −2 doubles the spread.

Ex. 4 (35 points)

a (5 points)
$z = (20 - 30)/8 = -1.25$ and $(44 - 30)/8 = 1.75$. Area from −1.25 to 0: $78.87/2 = 39.44\%$. Area from 0 to 1.75: $91.99/2 = 46.00\%$. Total **85.43%**.

b (5 points)
The quartiles leave 50% in the middle: $z = \pm 0.67$. IQR $= 2(0.67)(8) =$ **10.72 minutes** (quartiles 24.64 and 35.36).

c (7 points)
10% above 45 means 80% in the middle, so $z = 1.28$: $45 = \mu + 1.28\sigma$. 20% below 20 means 60% in the middle, so $z = -0.84$: $20 = \mu - 0.84\sigma$. Subtract: $25 = 2.12\sigma$, so $\sigma \approx 11.79$ minutes and $\mu = 45 - 1.28(11.79) \approx$ **29.91** minutes.

d (6 points)
Markov (scores ≥ 0): fraction ≥ 90 is at most $72/90 =$ **80%**. Chebyshev: 90 is $(90 - 72)/12 = 1.5$ SDs above the mean, so at most $1/1.5^2 =$ **44.44%** are that far from the mean in either direction, and so at most that many score 90 or above. Chebyshev is far better here: it uses the spread, while Markov uses only the mean. Both bounds are valid; take the smaller.

e (7 points)
(i) Let $Y = 100 - \text{score}$. Then $Y \geq 0$ with mean 28, and score ≤ 40 exactly when $Y \geq 60$. Markov: at most $28/60 = \mathbf{46.67\%}$. Attained when 7/15 of students score 40 and 8/15 score 100: mean $= (7 \cdot 40 + 8 \cdot 100)/15 = 72$.
(ii) 40 is $(72 - 40)/12 = 2.67$ SDs below the mean. Chebyshev: at most $1/(8/3)^2 = 9/64 = \mathbf{14.06\%}$.

> This is the "tricky" Review Question 4.13: Markov needs a non-negative quantity, so flip the scale.

f (5 points)
Below 48 is at least $(72 - 48)/12 = 2$ SDs below the mean. Above 99 is at least $(99 - 72)/12 = 2.25$ SDs above, which is also at least 2 SDs away. So both groups lie 2 or more SDs from the mean, and Chebyshev gives at most $1/2^2 = \mathbf{25\%}$. With an uneven interval, use the side that is *closer* to the mean.

Ex. 5 (30 points)

a (8 points)
$\bar{x} = 3$, $\bar{y} = 7$. Deviations: x: −2, −1, 0, 1, 2; y: −3, −1, −2, 3, 3.
$\text{SD}(x) = \sqrt{(4 + 1 + 0 + 1 + 4)/5} = \sqrt{2} = 1.414$; $\text{SD}(y) = \sqrt{(9 + 1 + 4 + 9 + 9)/5} = \sqrt{6.4} = 2.530$.
$\text{cov} = (6 + 1 + 0 + 3 + 6)/5 = 3.2$. $r = 3.2/(1.414 \times 2.530) = \mathbf{0.89}$. (Or average the products of standard units.)

b (5 points)
Adding a constant or multiplying by a positive constant leaves r alone; multiplying y by −3 flips the sign. New $r = \mathbf{-0.89}$. Covariance scales by both factors: $2 \times (-3) \times 3.2 = \mathbf{-19.2}$.

c (7 points)
$r = 0$ exactly when $\text{cov} = \overline{xy} - \bar{x}\bar{y} = 0$. For the first five points, $\sum xy = 4 + 12 + 15 + 40 + 50 = 121$. With the sixth point: $\overline{xy} = (121 + 6k)/6$, $\bar{x} = 21/6 = 3.5$, $\bar{y} = (35 + k)/6$. Setting them equal: $121 + 6k = 3.5(35 + k) = 122.5 + 3.5k$, so $2.5k = 1.5$ and $k = 0.6$.

d (5 points)
This is an **ecological correlation**, computed from state averages. Averaging removes the large variation among individuals within each state, so the points cluster tightly around a line. The individual-level correlation is typically much smaller than 0.85. (It also says nothing about causation.)

e (5 points)
Plot 1: 0.6. Plot 2: 0 (a strong but *nonlinear* U-shaped association; r measures only linear association). Plot 3: −0.85. Plot 4: −0.3. (Actual correlations of the plotted points: 0.6, 0.00, -0.85, -0.3.)

Ex. 6 (40 points)

a (6 points)
Slope $= r \cdot SD_y/SD_x = 0.6 \times 15/10 = 0.9$. The line passes through $(80, 70)$: $y - 70 = 0.9(x - 80)$, so $\hat{y} = 0.9x - 2$.

b (10 points)
$z_x = (90 - 80)/10 = 1$, so the predicted $z_y = 0.6$ and the predicted average is $70 + 0.6(15) =$ **79**.
RMS error $= \sqrt{1 - 0.6^2} \times 15 = 0.8 \times 15 = 12$. In this vertical strip, finals are roughly normal with mean 79 and SD 12. For 70: $z = (70 - 79)/12 = -0.75$. Area within 0.75 is 54.67%, so the left tail is $(100 - 54.67)/2 =$ **22.67%**.

c (10 points)
10th percentile: 80% in the middle, $z = -1.28$. Predicted $z_y = 0.6(-1.28) = -0.768$; predicted final $= 70 - 0.768(15) =$ **58.48**.
The final scores are football-shaped (normal), so the median equals the mean, 70. In the vertical strip: $z = (70 - 58.48)/12 = 0.96$. Area within 0.96 $\approx 66.3\%$, so the chance is about $(100 - 66.3)/2 \approx$ **17%**.

d (8 points)
Use the regression line for x on y: slope $= 0.6 \times 10/15 = 0.4$, through $(70, 80)$: $\hat{x} = 80 + 0.4(y - 70) = 0.4y + 52$. For $y = 100$: **92**.
Solving $y = 0.9x - 2$ for x gives $x = 113.3$, above the maximum possible score. The line in a) estimates the average y within a *vertical* strip; predicting x needs the average x within a *horizontal* strip. There are two regression lines, and each regresses toward its own mean.

e (6 points)
This is the **regression fallacy**. Since $r = 0.6 < 1$, the predicted $z_y = 0.6\, z_x$: students who are extreme on homework are, on average, less extreme on the final. The bottom group "improves" and the top group "falls back" purely because of scatter (luck, good and bad days), not because of anything about the final. The same pattern would appear in reverse if we sorted students by their final scores.